

Coding & Solution

Date	3 July 2026
Project Name	Rising Waters - Real-time Flood Prediction and Alert System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/example/rising-waters-flood-prediction
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Frameworks(s) Used	Flask (Python), scikit-learn, pandas
Key Features Implemented	- Machine learning-based flood risk predictive model. - Interactive map interface showing live station metrics. - Twilio-based immediate automated SMS alerting system.
Pending / Incomplete Features	None. All baseline operational criteria have been fulfilled.
Setup / Run Instructions	1. Clone the repository and run <code>`pip install -r requirements.txt`</code>. 2. Add API environment variables for Twilio and OpenWeather. 3. Launch the environment using command: <code>`python app.py`</code>.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application architecture meets all modular code benchmarks. Core calculations are separated from web-routing endpoints to ensure maintenance clarity. All local operational checks completed successfully without fatal exceptions, and automated priority warning outputs match anticipated thresholds.
